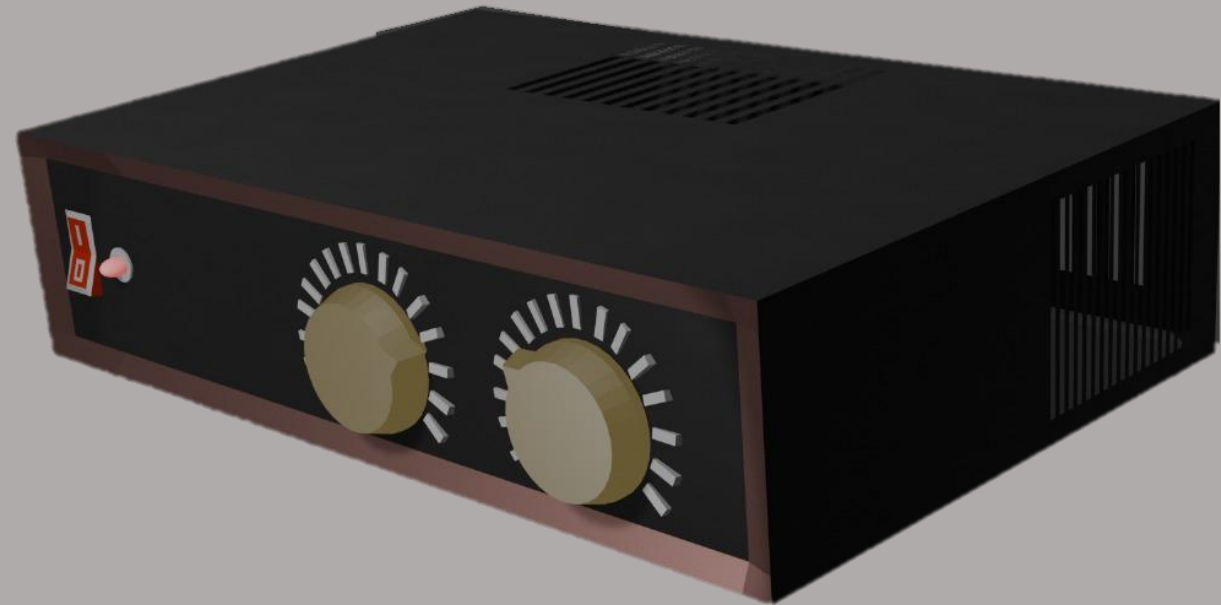


AUTOMATED SOUND REGULATOR

Skynet Tronic

- 230058N
- 230563H
- 230680M
- 230689A



PROBLEM

- Differences in voice levels or distance from the microphone require manual adjustments for a consistent output. **Consistency and comfortability.** Without assistance, overamplification can causes hearing damages
- Fluctuates in sound levels leads to **less engagement** of the audience. (lectures, events, public addresses)
- Many audio amplifiers apply a fixed gain regardless the output conditions



OUR SOLUTION

To address the issue of fluctuating audio levels that impact communication clarity, we propose the development of an external volume adjustment device.

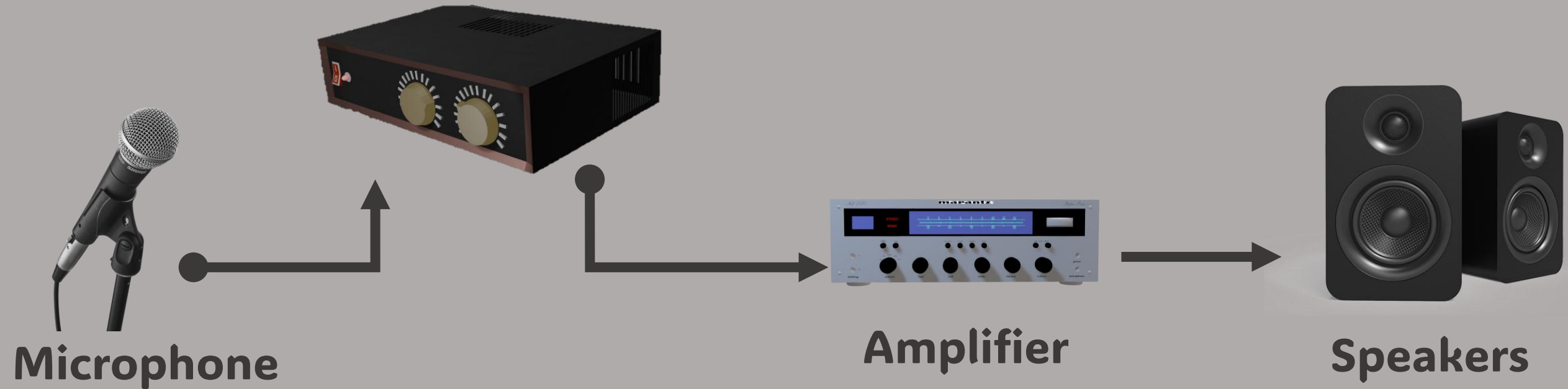
This system intelligently monitors and adjusts the audio output from an existing amplifier without needing to replace or heavily modify the current setup.



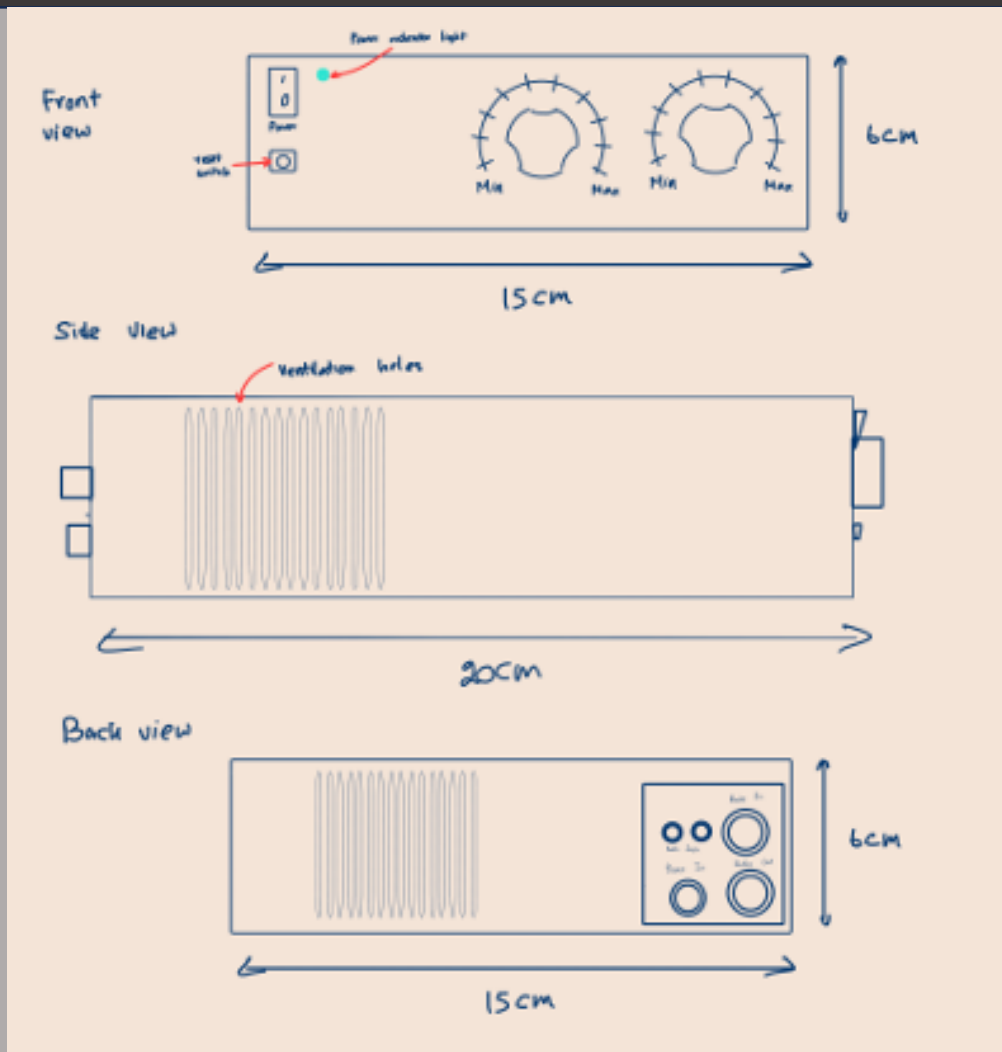
Automatic Sound Regulator

OUR SOLUTION

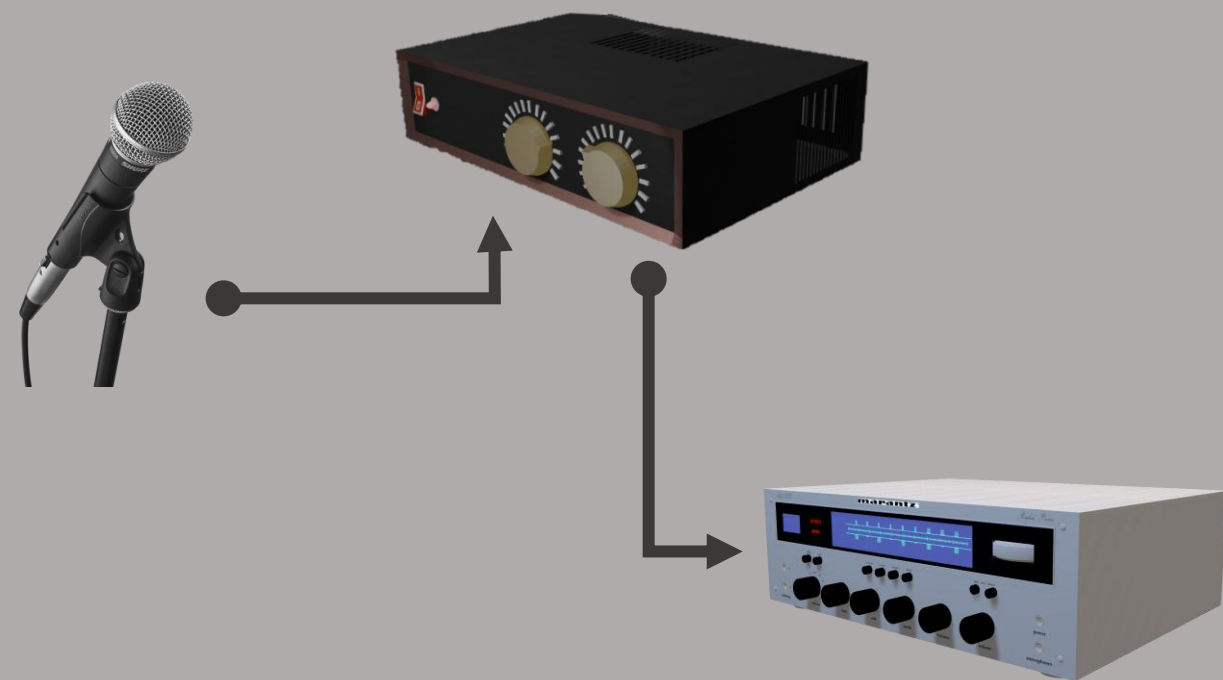
Automatic Sound Regulator



OUR SOLUTION



UI and UX



STAKE-HOLDER REVIEW

STAKE-HOLDERS

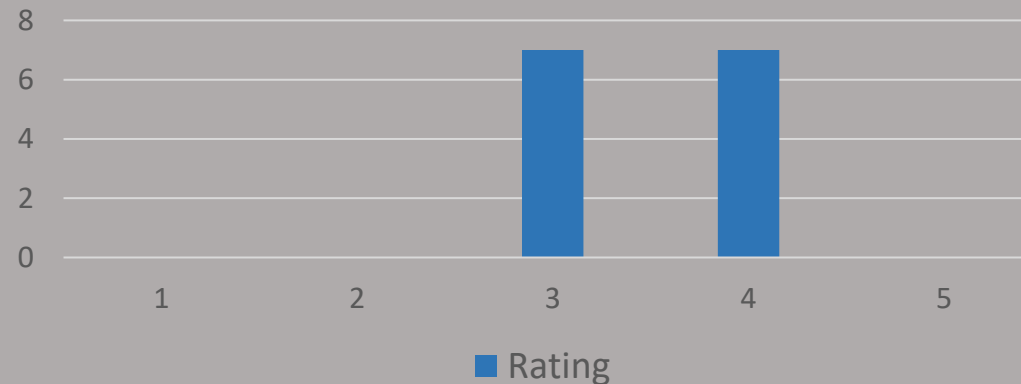
- Speakers / presenters
- End User (audience)
E.g. : students
- Institutions / Event Organizers
- Audio control personal / Engineers



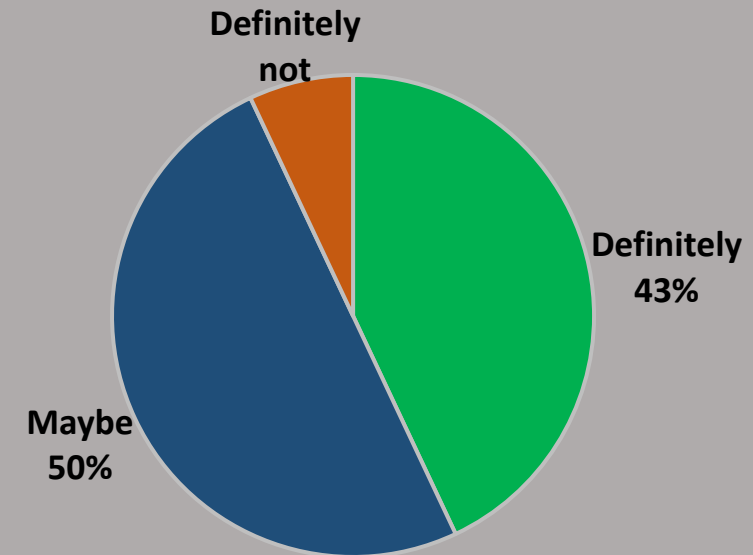
For more details

STAKE-HOLDER FEEDBACK SUMMARY

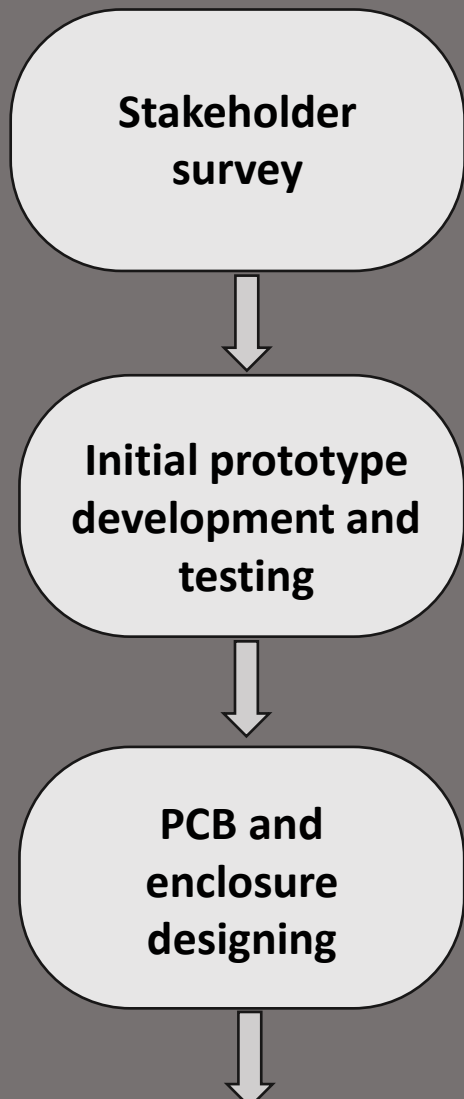
Existence of the problem and the necessity for solution



Feedback on our solution



Project Progress So Far



Feedback Survey: Automatic Volume Adjustment System for Speakers

Hello!

We are a group of students developing an **Automatic Volume Adjustment System** that intelligently manages speaker output.

The system uses a strategically placed sensor to detect the surrounding sound pressure level and sends data to a processor, which then decides whether to **amplify or reduce** the volume. This ensures that the speaker output stays within an optimal audible range — neither too loud nor too soft — based on the environment.

Our goal is to enhance audio comfort in varying acoustic conditions.

We would greatly appreciate your feedback on the **problem**, our **proposed solution**, and any **suggestions** you may have.

Questions Responses 14 Settings

Your Thoughts on the Solution

What are your initial thoughts on a system that automatically adjusts speaker volume based on environmental noise levels?

8 responses

I think it will be very useful specially for sensitive sound environments like study spaces, lecture halls and offices.

Its great , because most of the times there were lack of volumes or spikes at the volumes.

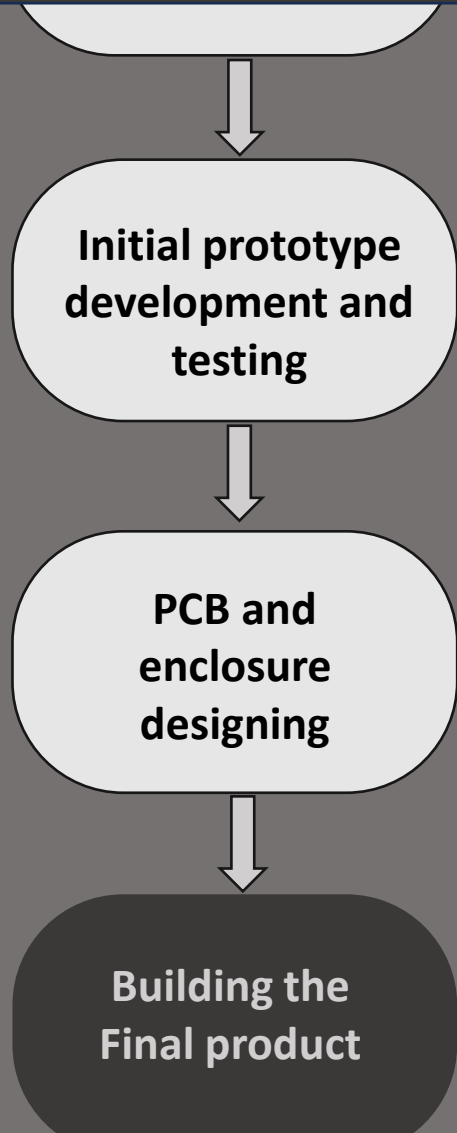
Feels like a crowded space with Apple (Apple TV has good voice enhancement, AirPods have great dynamic volume, I believe Sony does too). For home theatre problem feels solved by Apple... though maybe you can license different tech. For audiophile listening, I absolutely would NOT want this because it would mess with the original recording and intentions of the mix engineer.

If it can adjust accordingly to the different music or the sound that plays, and according to the environment the speaker is placed(eg: inside a small room, outside, big hall) that would be a great idea for modern speakers.

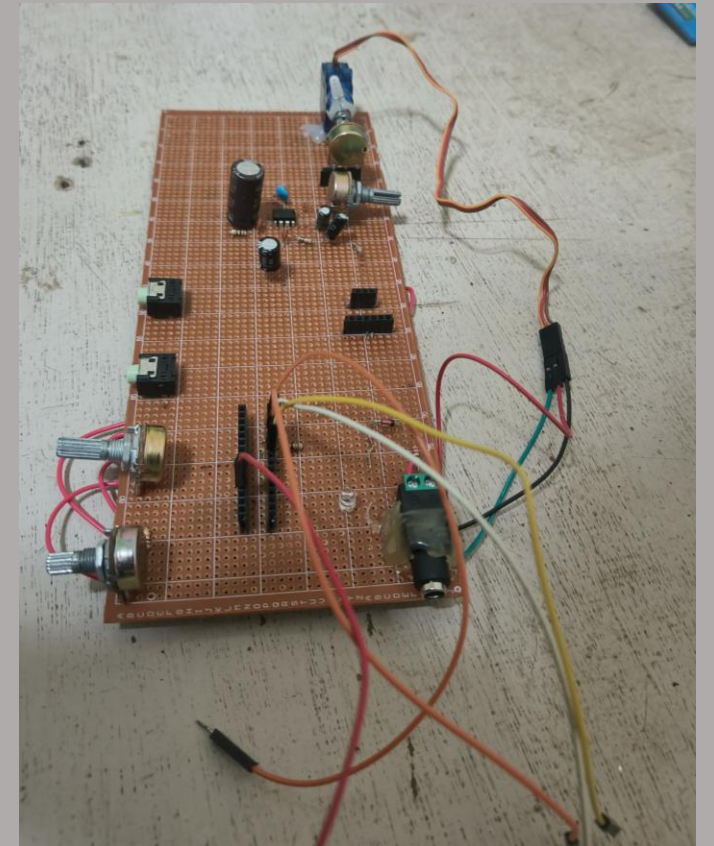
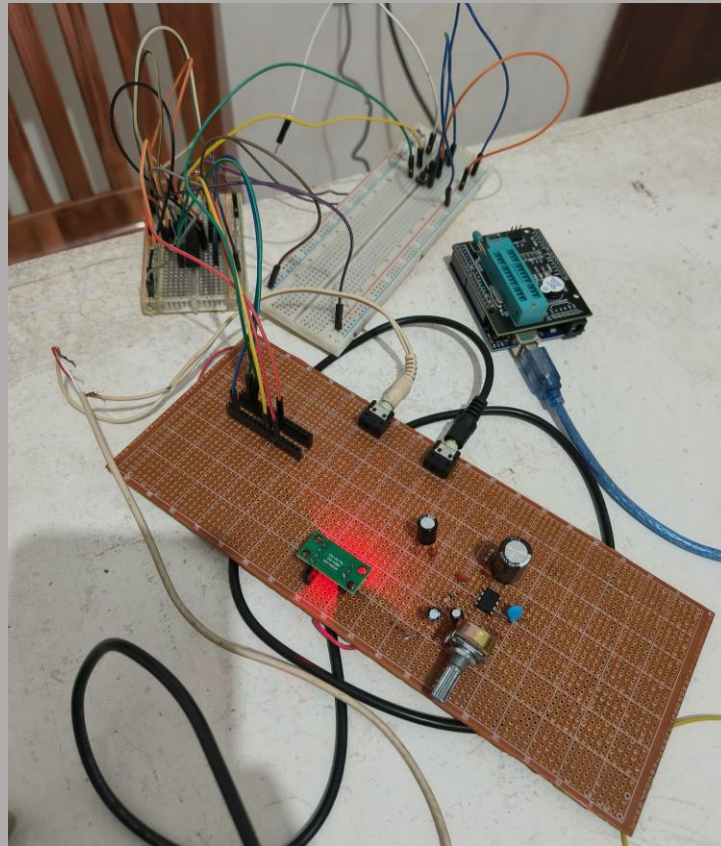
Need a lot of work and training for a Network to adjust sound in real time.

This is a very good solution idea for a practical problem we all are facing everyday, But i can't actually

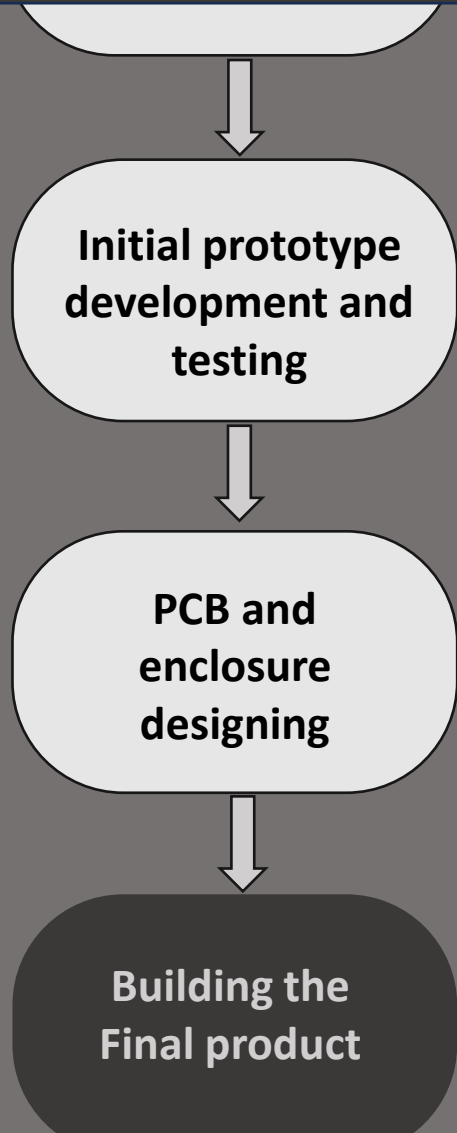
Project Progress So Far



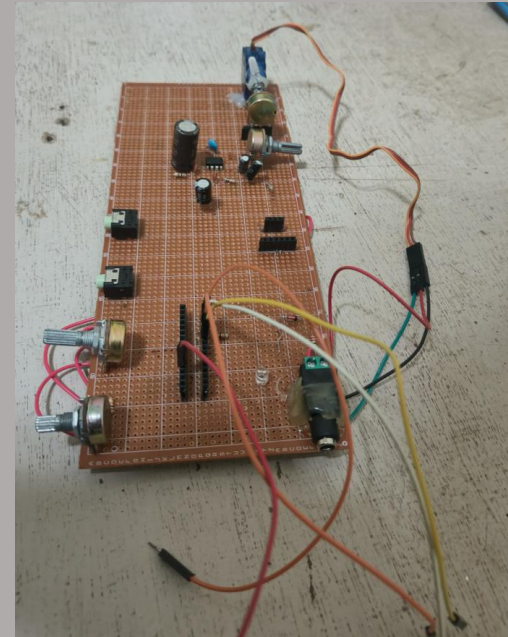
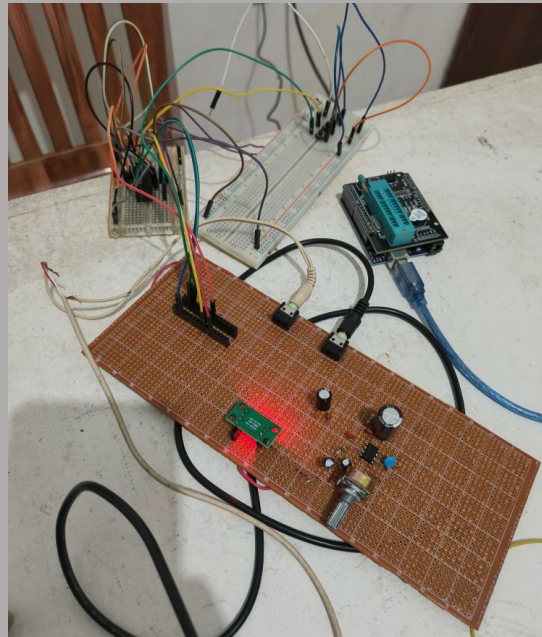
Testing Prototype



Project Progress So Far



Testing Prototype



Challenges we faced

- Digital potentiometer limitations
- Microphone based signal monitoring

Project Progress So Far

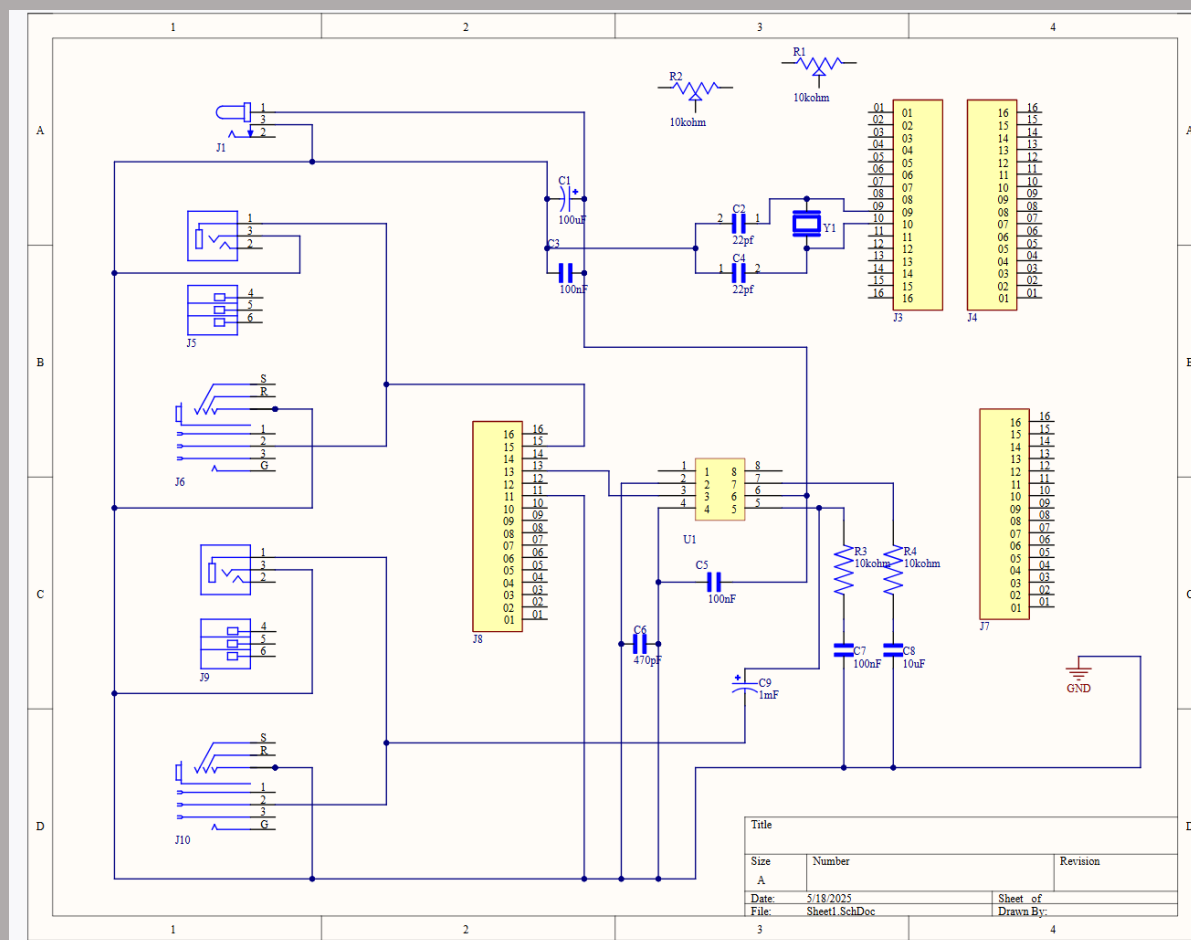
Initial prototype
development and
testing

PCB and
enclosure
designing

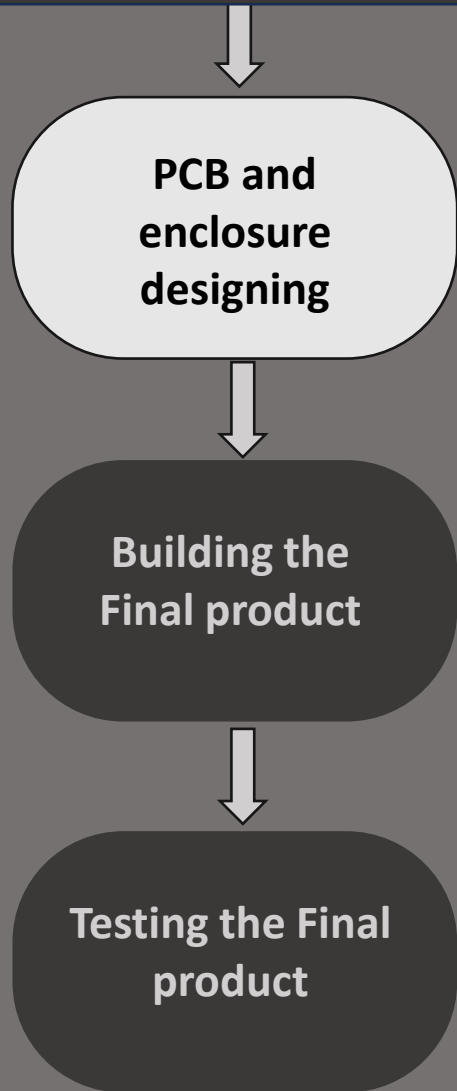
Building the
Final product

Testing the Final

Schematic Design



Project Progress So Far



To be done...

Thank you